



[Home](#) > [Science 6: Section 2](#) > [Topic 7 - Energy in the Atmosphere and Ocean](#) > [Topic 7 Test](#) > **Attempt 1**

Topic 7 Test

ATTEMPT SCORE

35.3333300000000004 / 38

01 Matching

3 / 3

Matching.

Match warm and cold surface water currents to their properties.

less dense	● — ●	A	WARM CURRENTS	✓
more dense	● — ●	B	COLD CURRENTS	✓
cool the air above them	● — ●	C	COLD CURRENTS	✓
warm the air above them	● — ●	D	WARM CURRENTS	✓
make the climate dry	● — ●	E	COLD CURRENTS	✓
make the climate wet	● — ●	F	WARM CURRENTS	✓

⚡ WARM CURRENTS

⚡ COLD CURRENTS

02 Short Answer/Essay

3 / 4

Greenland Solar Energy Reflection, 2014



SOURCE: National Snow & Ice Data Center

Short answer.

The graph shows how much solar radiation would be reflected from the land on the island of Greenland, situated in the far North Atlantic, through the year. Although Greenland is mostly covered with a year-round glacial ice cap, the edges of the island can lose their snow cover and the reflectiveness of the surface of the snow pack changes when melting occurs.

1. Explain the pattern of the graph and what effect it will have on the local climate.
2. Will warmer average global temperatures encourage the effect seen in the graph or reduce it? Why?

1. During the Winter snow accumulates on the ice caps. And since snow is reflective, the reflectiveness increases. The same happened with Fall. Summer is a different story. Summer is at least a few degrees hotter than Winter and Fall. So in this case, the snow melted and the glacial ice cap is revealed. And since ice isn't that reflective, the percentage lowered to around 70 degrees. The local climate would be "hot" in the Summer and cold during Winter, Spring, and Fall.

2. Warmer global temperatures will reduce the effect seen in the graph. When global temperature heats up the seasons will too. So Winter, Spring, Summer, and Fall will all be hotter than before the rise in temperature. This means that more of the snow covered glacial ice caps will melt revealing the ice-cap. Lowering the amount of reflectiveness sent back into the atmosphere

146 / 250 Word Limit

03 Multiple Choice

2 / 2

Choose the best answer.

What is the major driving force that causes land and sea breezes?

- ☐ global wind belts
- ☒ heating and cooling of water and land
- ☐ curved surface of Earth
- ☐ rotation of the Earth on its axis

**04 True/False****2 / 2****True or false.**

Greenhouse gases reflect energy, leading to warming of the atmosphere.

- ☐ True
- ☒ False

**Student's Correction**

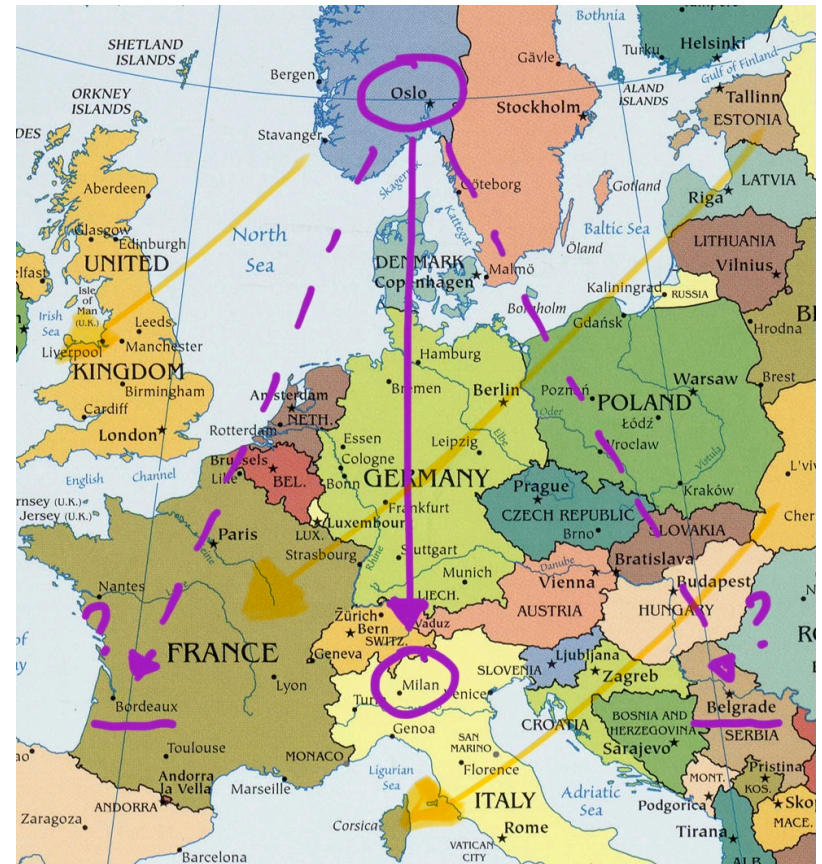
Greenhouse gases absorb the reflected energy from the surface, leading to the warming of the atmosphere.

Correction Guidelines**05 Multiple Choice****2 / 2**

Choose the best answer.

The Norwegian cross country skiing team flies from Norway's capital of Oslo to Milan, where the Olympics are held. Milan is due south of Oslo, but one of the younger athletes notices on his compass that the plane is flying south east. The flight attendant explains that the Earth is still spinning while the plane is in the air, so if the plane were to fly due south, the Coriolis effect (helpfully marked with the yellow arrows) would make them land in the wrong country!

Where does the flight attendant say the plane would land if the plane flew due south?

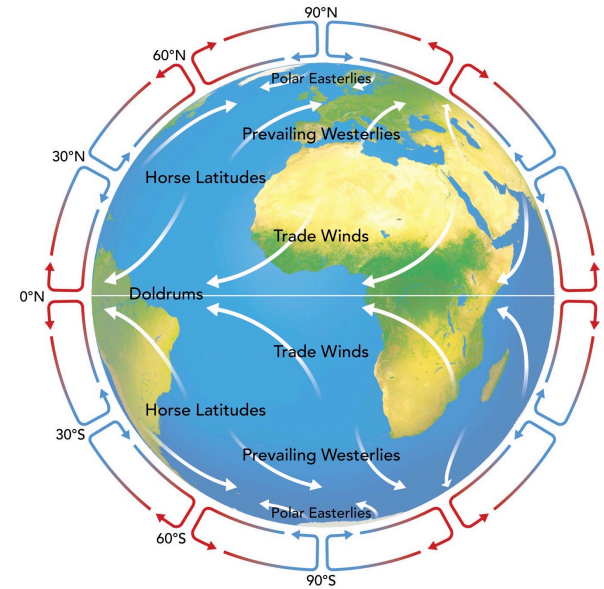


- ☐ Milan, Italy
- ☐ Oslo, Norway
- ☒ Bordeaux, France
- ☐ Belgrade, Serbia



Fill in the blanks.

The doldrums are sometimes called a "windless zone" because sailing ships often become stranded in these areas. Air in the doldrums rises because it is 1 warmed by the sun at the Equator ✓. After the air rises, 2 it begins to cool off and sink ✓.

**07 Multiple Choice**

— / —

Choose the best answer.

What is the fate of the light energy that is absorbed by Earth's surface?

- ☐ a small amount is converted to chemical energy by plants, the rest is converted to heat and radiated back ✓
- ☒ a small amount is converted to heat and radiated back, the rest is converted to chemical energy by plants ✗
- ☐ a small amount is converted to chemical energy by plants, the rest is converted to wind energy
- ☐ a small amount is converted to chemical energy by plants, the rest is converted to heat and conducted to Earth's core

08 Multiple Choice

2 / 2

Choose the best answer.

Which explains why global winds do not move in straight paths?

- ☐ air pressure
- ☒ Earth's rotation
- ☐ warm air rises
- ☐ jet streams

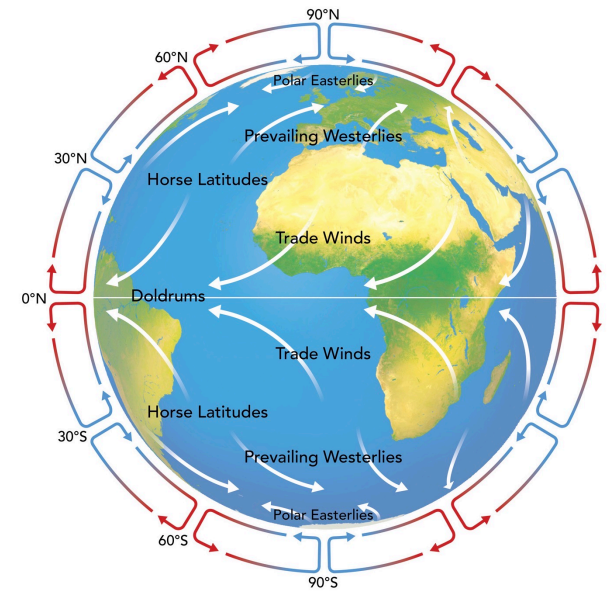


09 Multiple Choice

1 / 2

Select all that apply.

The equator sees large amounts of rainfall throughout much of the year because the rising air cools and its moisture condenses and falls as rain. What climate conditions do you expect at 30 north and south?


☒ strong Westerly winds

☐ strong Easterly winds

☒ very litte precipitation

☐ weak winds

☐ heavy precipitation

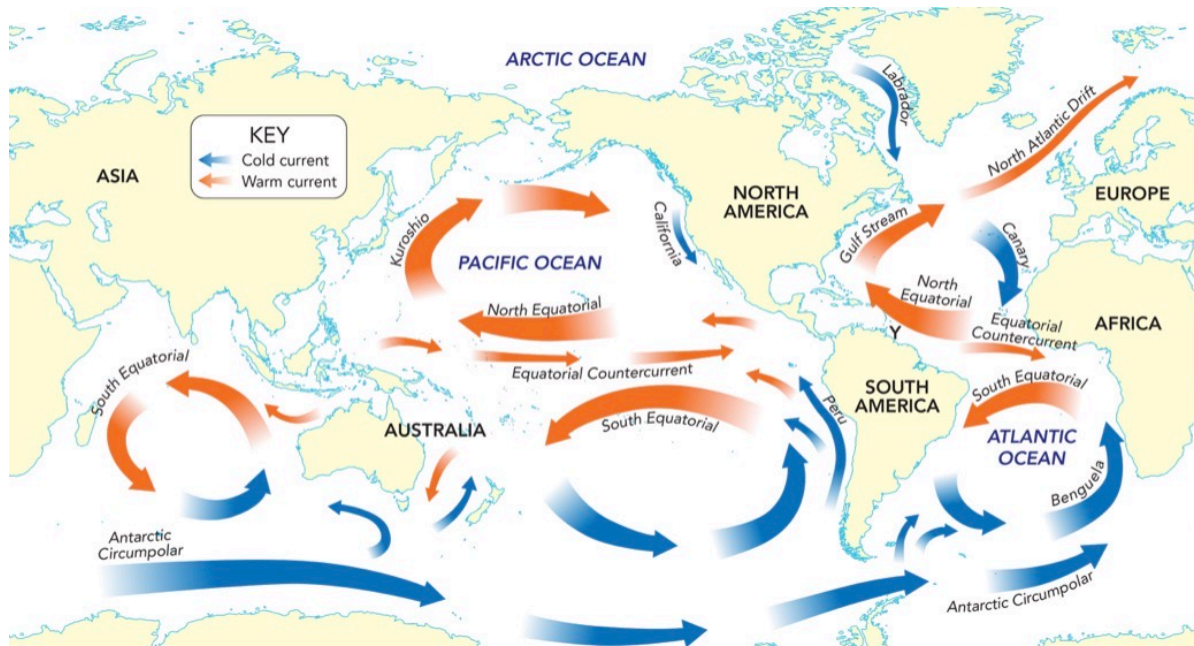
✗

✓

✓

10 Multiple Choice

2 / 2



Choose the best answer.

Which is **not true** about global ocean currents?

- ☒ ocean currents travel against global wind patterns
- ☐ water cools near the poles before traveling back towards the Equator
- ☐ currents travel west in the tropics (but east at the Equator) due to the Coriolis effect
- ☐ the western side of land masses experience less humidity due to the colder currents



11 Matching

5 / 5

Matching.

Match the example to the vocabulary term. *Not all terms will be used!*

Anemometer

A

a device that measures wind speed ✓

Horse Latitudes

B

better hope your boat has a motor if you're here, because the winds aren't strong enough to be any help ✓

Jet Stream

C

these winds travel upwards of 110mph -- but don't worry, because they do so at an altitude of 10km ✓

El Niño

D

this weather event makes the northwestern coast of North America unusually hot ✓

Radiation

E

solar energy reaches Earth this way ✓

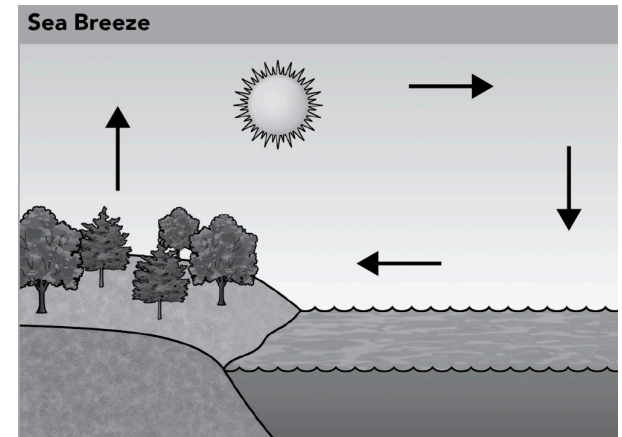
❖ European explorers in the 15th century used these to get to the Americas

❖ it flows away from the Equator towards the poles in a familiar pattern

12 Ordering**1.33 / 2**

Ordering.

Order the steps of the process that causes sea breezes.



the sun rises



solar radiation heats the land



the land heats the air, which then rises



air high above the land moves over the sea to replace sea air



4 **Correct answer:** air over the water flows toward land

upper air descends to replace air over the water



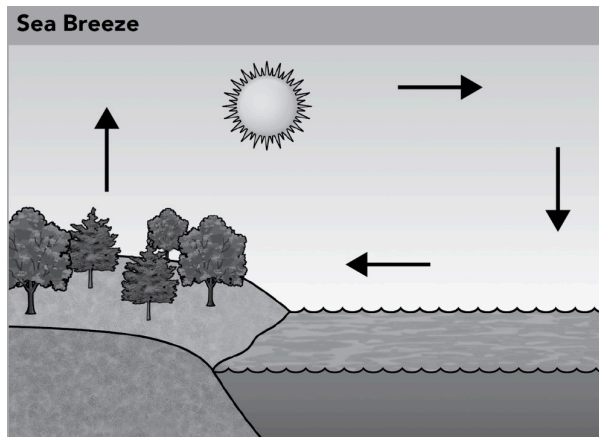
air over the water flows toward land



6 **Correct answer:** air high above the land moves over the sea to replace sea air

13 Short Answer/Essay

4 / 4

**Short answer.**

The movement of air masses occurs because of density differences. Explain the role of density in the movements shown in the diagram.

When the air above the land heats up, the density lowers. The low density air rises because of convection currents. Then because the air above the sea has a higher density, the low density air moves toward the high density, cold air. then the low density air cools down and sinks. Replacing the cold, high dense air above the sea. Then the cold, dense air moves toward the warm land because of convection currents. This is the role of density shown in the diagram.

84 / 100 Word Limit

14 Multiple Choice

2 / 2

Choose the best answer.

Farmer McDouglas looks at the wind vane on his barn and sees the rooster pointing north, showing that the wind is blowing south to north. What would a wind blowing this direction be called?



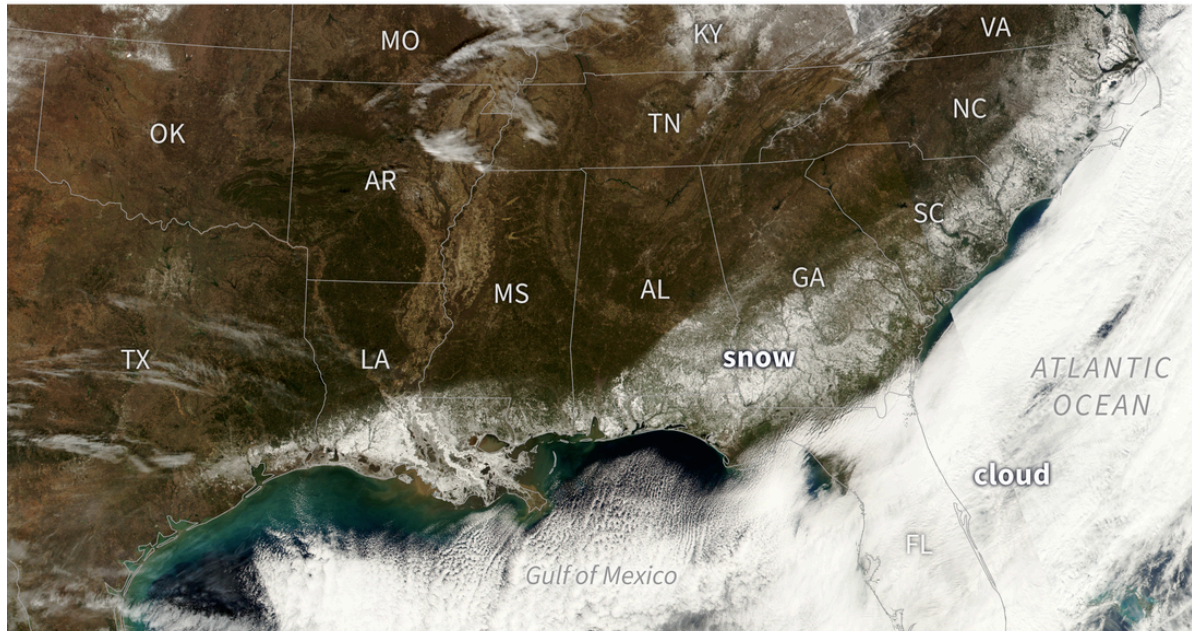
- ☐ doldrums
- ☒ southerly
- ☐ northerly
- ☐ prevailing

**15** Fill in the Blank Drag and Drop

2 / 2

Fill in the blanks.

Snowfall in the Southeastern U.S.



January 22, 2025

NOAA Climate.gov
Data: VIIRS/NOAA-21

In January of 2025, the southern United States experienced historic levels of snow. 1 Cold air ✓ from the Arctic was brought south by the jet stream, causing cold temperatures all over the continental US. 2 Warm, moist air ✓ from the Gulf of Mexico met with the continental polar air masses over coastal states like 3 Louisiana and ✓ic, causing the water vapor to condense and then precipitate. Landlocked states like 4 Arkansas and ✓er saw little snowfall, despite the freezing temperatures, because the continental polar air was relatively dry.

16 Multiple Choice

2 / 2

Choose the best answer.

Through absorption and reflection of energy from the sun, the Earth's atmosphere surface keep the planet within habitable temperatures. Which of the following statements is **not true**?

- ☐ about 25% of energy from the sun reflects off the atmosphere before ever hitting the surface
- ☐ energy absorbed by the surface heats the surface and the air above it
- ☐ an increase in greenhouse gases will lead to warmer global temperatures
- ☒ the polar ice caps absorb more energy than any other part of the planet

